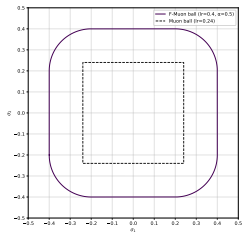
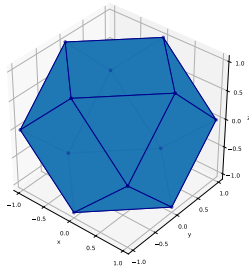


The Ky Fan Norms and Beyond: Dual Norms and Combinations for Matrix Optimization

Alexey Kravatskiy

MIPT

December 20, 2025



Moving beyond the spectral norm and Muon

F-Fanions: Muon, Neon, NSGD, Dion without EF, and so much more

From $\|\cdot\|_{\text{op}}$ and Muon:

$$\mathbf{X}^{t+1} = \mathbf{X}^t - \eta UV^\top$$

To $\|\cdot\|_{\text{KF-k}}^\dagger$ and a general F-Fanion:

$$\mathbf{X}^{t+1} = \mathbf{X}^t - \eta \left(\alpha \sum_{i=1}^k u_i v_i^\top + (1 - \alpha) \frac{\mathbf{M}^t}{\|\mathbf{M}^t\|_{\text{F}}} \right), \alpha \in [0, 1]$$

To $\|\cdot\|_{\text{C}^\dagger - \text{KF-k}}^\dagger$ and a general S-Fanion:

$$\mathbf{X}^{t+1} = \mathbf{X}^t - \gamma_t \left(\alpha \sum_{i=1}^k \mathbf{u}_i \mathbf{v}_i^\top + (1 - \alpha) \eta \text{sign}(\mathbf{M}^t) \right) \alpha \in [0, 1], \eta > 0$$

Linear Minimization Oracle (LMO)

Let us equip $\mathbb{R}^{m \times n}$ with a norm $\|\cdot\|$. Its dual is $\|\mathbf{X}\|^\dagger = \sup_{\|\mathbf{X}'\| \leq 1} \langle \mathbf{X}, \mathbf{X}' \rangle$. $\langle \cdot, \cdot \rangle$ is the Frobenius product.

LMO leads to the update

$$\mathbf{X}^{t+1} = \mathbf{X}^t - \eta \arg \max_{\mathbf{X} \in \mathcal{B}_1} \langle \mathbf{M}^t, \mathbf{X} \rangle = \mathbf{X}^t - \eta \{ \Delta \in \mathcal{B}_1 \mid \langle \mathbf{M}^t, \Delta \rangle = \|\mathbf{M}^t\|^\dagger \}$$

Recipe: It means we seek Δ from the 1-norm ball that delivers $\langle \mathbf{M}^t, \Delta \rangle = \|\mathbf{M}^t\|^\dagger$

We will often utilize the SVD of $\mathbf{M}^t$: $\mathbf{M}^t = \mathbf{U}\Sigma\mathbf{V}^\top$.

Of Matrix and Vector Algorithms

Table: lmo optimizers in Schatten S_p norms and in l_p norms.

Method	lmo constraint set $\mathcal{D}$	lmo	Reference
Normalized SGD	l_2 -ball, S_2 -ball	$-\eta \frac{g}{\ g\ _2} = -\eta \frac{g}{\ g\ _F}$	(Hazan et al., 2015)
Momentum Normalized SGD	Ball in l_2 , or Ball in S_2	$-\eta \frac{g}{\ g\ _2} = -\eta \frac{g}{\ g\ _F}$	(Cutkosky et al., 2020)
SignSGD	Ball in Max-norm l_∞	$-\eta \text{sign}(g)$	(Bernstein et al., 2018, Thm. 1)
Signum	Ball in Max-norm l_∞	$-\eta \text{sign}(g)$	(Bernstein et al., 2018, Thm. 3)
Muon	Ball in Spectral S_∞	$-\eta UV^\top$	(Jordan et al., 2024)
Gauss-Southwell Coordinate Descent	Ball in l_1	$-\eta \sum_{i \in \arg \max g_i^t } \text{sign}(g_i^t) e_i$	(Shi et al., 2016, p.19)
Neon	Ball in Nuclear S_1	$-\eta u_1 v_1^\top$	(Kravatskiy et al., 2025) (Pethick et al., 2025)
ν SAM			

The rank gap problem

Schatten p -norms $\|\mathbf{M}^t\|_{S_p} = \left(\sum_{i=1}^{\min(m,n)} \sigma_i^p\right)^{1/p}$ produce the update

$$\mathbf{X}^{t+1} = \mathbf{X}^t - \gamma_t \mathbf{U} \frac{\text{diag}\left(\sigma_1^{q-1}, \dots, \sigma_{\min(m,n)}^{q-1}\right)}{\left(\sum_{i=1}^{\min(m,n)} \sigma_i^q\right)^{\frac{q-1}{q}}} \mathbf{V}^\top$$

where p and q satisfy $p^{-1} + q^{-1} = 1$ (Cesista, 2025).

Ky Fan k -norms $\|\mathbf{M}^t\|_{\text{KF-}k} = \sum_{i=1}^k \sigma_i$ produce either

$$\mathbf{X}^{t+1} = \mathbf{X}^t - \gamma_t \mathbf{u}_1 \mathbf{v}_1^\top \text{ or } \mathbf{X}^{t+1} = \mathbf{X}^t - \frac{\gamma_t}{k} \mathbf{U} \mathbf{V}^\top,$$

that is either Neon or Muon (Kravatskiy et al., 2025).

Thus, the updates are either one-rank or full-rank.

Ky Fan k -rank $\|\mathbf{M}^k\|_{\text{KF}-k}^\dagger$ and Fanions

Deriving Fanions by Recipe

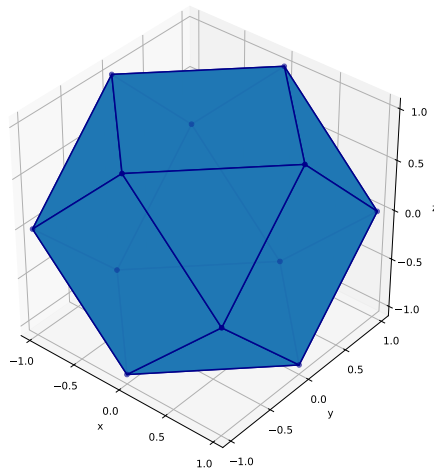
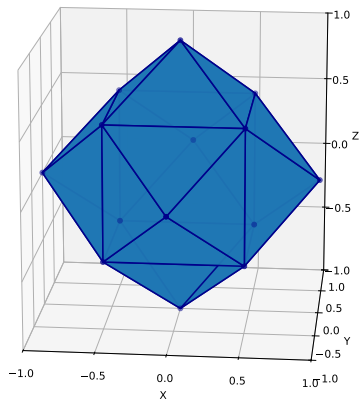
$\|\mathbf{M}^t\|_{\text{KF}-k}^{\dagger\dagger} = \|\mathbf{M}^t\|_{\text{KF}-k}$, and $\Delta = \sum_{i=1}^k u_i v_i^\top$ with
 $\|\Delta\|_{\text{KF}-k}^\dagger = \max\{\frac{1}{k}\|\Delta\|_{\text{nuc}}, \|\Delta\|_{\text{op}}\} = \max\{\frac{1}{k}k, 1\} = 1$ delivers it:

$$\langle \mathbf{M}^t, \Delta \rangle = \langle \mathbf{U} \Sigma \mathbf{V}^\top, \sum_{i=1}^k u_i v_i^\top \rangle = \sum_{i,j=1}^{r,k} \langle u_i \sigma_i v_i^\top, u_j v_j^\top \rangle = \sum_{i=1}^k \sigma_i = \|\mathbf{M}^t\|_{\text{KF}-k}$$

Hence,

$$\mathbf{X}^{t+1} = \mathbf{X}^t - \eta \sum_{i=1}^k u_i v_i^\top$$

Geometric point of view: Ky Fan norms



Ky Fan rank-2 ball and its dual in a 3D singular-value space

Balls of duals to convex combinations of norms

Lemma

Let $\|\cdot\|_{(1)}, \dots, \|\cdot\|_{(n)}$ be norms on a finite-dimensional Euclidean space, and let $\alpha_1, \dots, \alpha_n$ be non-negative reals. Define

$$\|\cdot\| := \sum_{i=1}^n \alpha_i \|\cdot\|_{(i)}.$$

Then the dual unit ball of $\|\cdot\|$ satisfies

$$\mathcal{B}_{\|\cdot\|}^\dagger = \sum_{i=1}^n \alpha_i \mathcal{B}_{\|\cdot\|_{(i)}}^\dagger,$$

where $\sum$ denotes the Minkowski sum and $\mathcal{B}_{\|\cdot\|_{(i)}}^\dagger$ is the unit ball of the dual norm $\|\cdot\|_{(i)}^\dagger$.

Conic combination of LMO algorithms is an LMO algorithm

Corollary

Let there be a finite family of LMO-based algorithms indexed by $i = 1, \dots, n$, where the $\mathbf{X}^{t+1}$ update in the i -th algorithm is defined by

$$\mathbf{X}^{t+1} - \mathbf{X}^t = \gamma_t \text{LMO}_i(\mathbf{M}^t),$$

and LMO_i corresponds to the unit ball of norm $\|\cdot\|_i$. For arbitrary non-negative $\alpha_1, \dots, \alpha_n$, the algorithm with the update given by

$$\mathbf{X}^{t+1} - \mathbf{X}^t = \gamma_t \sum_{i=1}^n \alpha_i \text{LMO}_{\|\cdot\|_{(i)}}(\mathbf{M}^t)$$

is an LMO-algorithm itself, with LMO corresponding to the unit ball of the norm $\|\cdot\|$ dual to the norm given by $\sum_{i=1}^n \alpha_i \|\cdot\|_{(i)}^\dagger$.

$\|M^k\|_{F-KF-k}^\dagger$ and F-Fanions

Deriving F-Fanions by Recipe

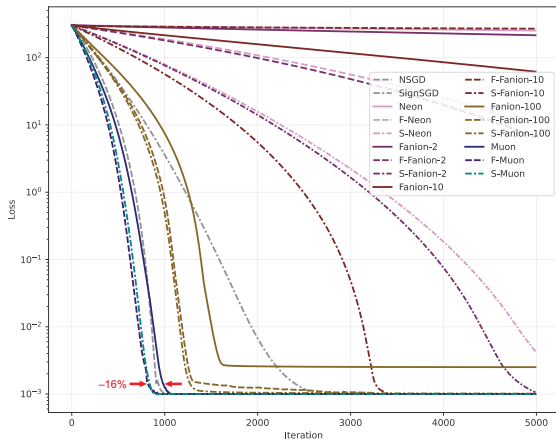
$\|M^t\|_{F-KF-k}^{\dagger\dagger} = \|M^t\|_{F-KF-k}$, and $\Delta = \alpha \sum_{i=1}^k u_i v_i^\top + (1 - \alpha) \frac{M^k}{\|M^k\|_F}$ delivers it:

- ① $\|\Delta\|_{F-KF-k}^\dagger \leq 1$ because $\sum_{i=1}^k u_i v_i^\top$ lies in $B_{\|\cdot\|_{KF-k}}$ and $\frac{M^k}{\|M^k\|_F}$ lies in $B_{\|\cdot\|_F}$, so Δ lies in the Minkowski sum
- ② $\langle M^t, \Delta \rangle = \alpha \langle U \Sigma V^\top, \sum_{i=1}^k u_i v_i^\top \rangle + (1 - \alpha) \langle M^t, \frac{M^t}{\|M^t\|_F} \rangle =$
 $\alpha \sum_{i=1}^k \sigma_i + (1 - \alpha) \|M^t\|_F = \|M^t\|_{F-KF-k}$

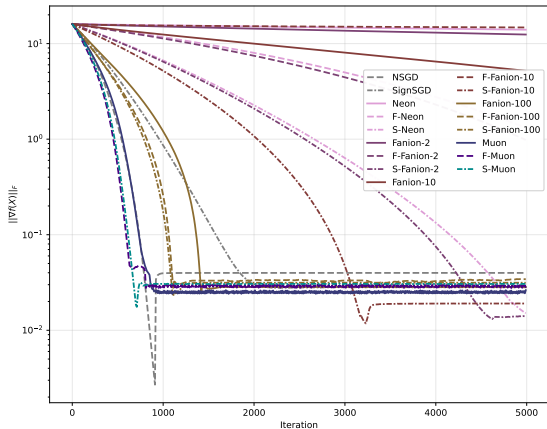
Hence,

$$X^{t+1} = X^t - \eta \left(\alpha \sum_{i=1}^k u_i v_i^\top + (1 - \alpha) \frac{M^t}{\|M^t\|_F} \right)$$

A smooth convex problem for a matrix



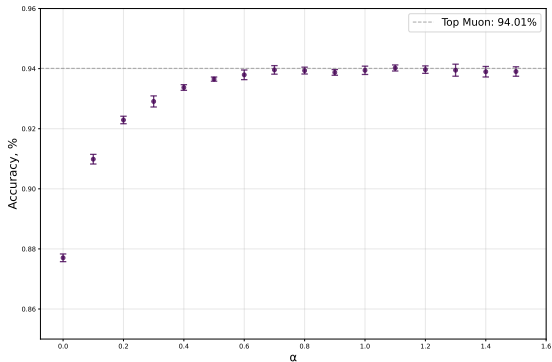
Loss vs iteration



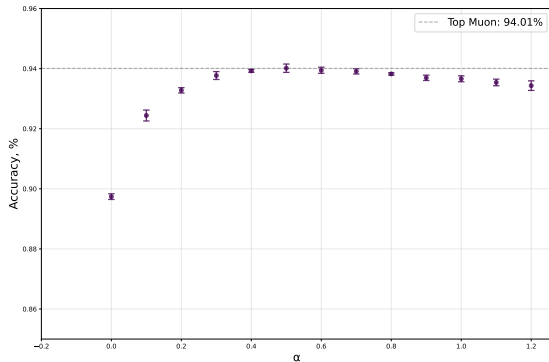
Frobenius norm of the gradient vs iteration

$$F(\mathbf{X}) = \frac{1}{2} \|\mathbf{M}^{1/2} \mathbf{X} \mathbf{N}^{1/2}\|_F^2 \rightarrow \min_{\mathbf{X} \in \mathbb{R}^{500 \times 500}}, \lambda(\mathbf{M}), \lambda(\mathbf{N}) \sim U[0, 1], \mathbf{X}_{ij}^0 \sim \mathcal{N}(0, 1)$$

F-Muon on CIFAR-10 airbench



With params tuned for Muon, as in (Keller, 2023)

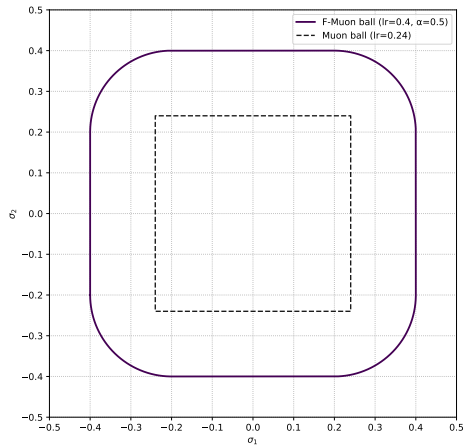


With params tuned for $\alpha = 0.5$ F-Muon

F-Muon with $\alpha = 0.5$ matches Muon after tuning

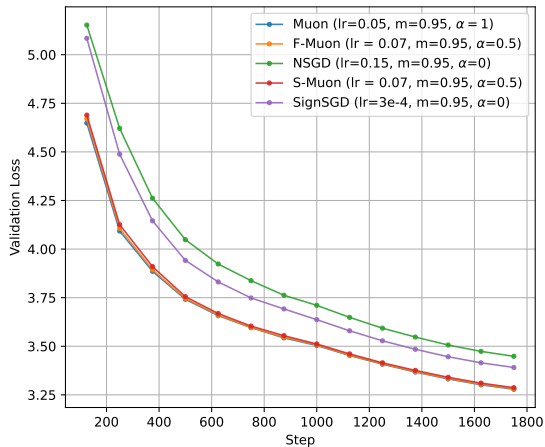
❗ F-Fanion with $\alpha > 1$ is not an LMO algorithm

LMO balls on CIFAR-10

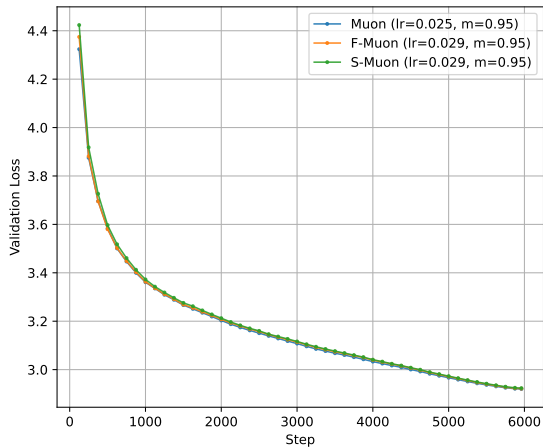


LMO balls for the learning rates from the experiments

NanoGPT and GPT-2 Medium



NanoGPT



GPT-2 Medium

Conclusion

- Muon, ν -SAM, and Dion without error feedback are Fanions
- A combination of LMO algorithms is an LMO algorithm
- F-Muon and S-Muon are on par with Muon

For the future: Estimating (L_0, L_1) smoothness for new norms, as in (Riabinin et al., 2025)

References I

-  Bernstein, Jeremy et al. (2018). “signSGD: Compressed optimisation for non-convex problems”. In: *International conference on machine learning*. PMLR, pp. 560–569.
-  Cesista, Franz Louis (Feb. 2025). *Steepest Descent Under Schatten- p Norms*. URL: <https://leloykun.github.io/ponder/steepest-descent-schatten-p/>.
-  Cutkosky, Ashok, Harsh Mehta, and Francesco Orabona (2020). “Momentum-based variance reduction in nonconvex SGD”. In: *Advances in Neural Information Processing Systems*.
-  Hazan, Elad, Kfir Levy, and Shai Shalev-Shwartz (2015). “Beyond convexity: Stochastic quasi-convex optimization”. In: *Advances in neural information processing systems* 28.
-  Jordan, Keller et al. (2024). *Muon: An optimizer for hidden layers in neural networks*. URL: <https://kellerjordan.github.io/posts/muon/>.
-  Keller, Jordan (2023). *cifar10-airbench*. URL: <https://github.com/KellerJordan/cifar10-airbench>.

References II

-  Kravatskiy, Alexey et al. (2025). *The Ky Fan Norms and Beyond: Dual Norms and Combinations for Matrix Optimization*. *arXiv: 2512.09678 [math.OC]*. URL: <https://arxiv.org/abs/2512.09678>.
-  Pethick, Thomas et al. (2025). “ ν SAM: Memory-Efficient Sharpness-Aware Minimization via Nuclear Norm Constraints”. In: *Transactions on Machine Learning Research*. ISSN: 2835-8856. URL: <https://openreview.net/forum?id=V6ia5hWIMD>.
-  Riabinin, Artem et al. (2025). “Gluon: Making Muon & Scion Great Again! (Bridging Theory and Practice of LMO-based Optimizers for LLMs)”. In: *arXiv preprint arXiv:2505.13416*.
-  Shi, Hao-Jun Michael et al. (2016). “A primer on coordinate descent algorithms”. In: *arXiv preprint arXiv:1610.00040*.